HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained reusable **UVM test benches** and verification environments for complex SoC subsystems, integrating **constrained random** and **formal verification** methodologies.
- Contributed to **power-aware verification** efforts for next-generation FPGA device products, ensuring functional correctness across various operating states.
- Collaborated with **architecture and design teams** to refine specifications and ensure seamless integration, aiming for first-silicon success on new chip designs.
- Authored **Python scripts** for automating data validation and test case generation, reducing manual effort in the verification cycle by approximately **30%**.
- Participated in **code reviews** for RTL and verification code, ensuring adherence to quality standards and best practices within the engineering team.
- Gained exposure to **ML-enabled verification** methodologies, exploring their application to optimize coverage and debug complex issues.
- Managed source code using **Git**, implementing branching strategies and participating in **Agile/Scrum ceremonies** for iterative development and delivery.

SKILLS

Verification Methodologies – UVM, Constrained Random, Formal Verification, Coverage-Driven Verification

Hardware Description Languages – SystemVerilog

Programming & Scripting – Python, Perl, TCL/TK, C/C++

SoC Architecture – CPU Subsystems, ARM, RISC-V, AMBA Bus Systems

Tools – Siemens EDA, Cadence Verification Suite, Synopsys Verification Suite, Git, Perforce, SVN

Domain – FPGA Verification, Power-Aware Verification, RTL Verification

PROJECTS

FPGA Subsystem Verification Automation

- Developed a suite of **Python scripts** to automate the generation of test vectors and validation checks for critical FPGA subsystems, integrating with existing **UVM environments**.
- Implemented checks for **power-aware verification** scenarios, simulating various low-power states and ensuring correct state transitions and wake-up sequences.
- Authored detailed documentation and collaborated with senior engineers to integrate automated checks into the main CI/CD pipeline, reducing regression time by **15%**.

SoC Bus Protocol Compliance Testing

- Designed and implemented **SystemVerilog** test scenarios to verify compliance of custom IP with AMBA bus protocols, focusing on transaction ordering and interrupt handling.
- Created reusable verification components within a **UVM framework** to accelerate the development of test benches for various bus interfaces.
- Executed extensive simulation runs and analyzed waveforms using **Siemens QuestaSim** to identify and debug protocol violations.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)